Network Security

pfSense Network Security Configuration

Introduction

pfSense is a firewall and routing system that is built on top of FreeBSD. The pfSense works as a firewall that looks at each packet of data. It also works as a router that helps figure out where data should travel. It works as a Network Address Translation gateway. All of these things are part of one free software package.

The pfSense is an operating system. So it gives people in charge a lot of control over how data packets are handled at a basic level. This means administrators can make specific rules about how data is allowed to travel. They can also set up VPN connections and other services, like DHCP and DNS. They can control how fast data travels without requiring special hardware. The pfSense system is very flexible.

In this practical, pfSense's core capabilities packet filtering, static and dynamic routing between network segments, and NAT for translating private LAN addressing to a routable WAN address form the foundation of a self contained edge security gateway suitable for a virtualized laboratory environment.

The purpose of this practical is to deploy a secure virtual environment using Oracle VirtualBox. By installing and configuring pfSense as a gateway, we establish a secure boundary between a simulated external network (WAN) and a private internal network (LAN) containing client machines.

Objectives

- Prepare a virtual environment for pfSense and a Windows client (Created an Ubuntu client since a Windows client upon the Windows 11 OS is too slow and it takes much time to do the configurations).
- Install pfSense and complete the initial console configuration.
- Configure the WAN interface using a static IPv4 address.
- Configure the LAN interface as the internal gateway.
- Install and configure a Windows/Ubuntu client on the pfSense LAN.
- Enable DHCP address assignment for LAN clients.
- Configure the pfSense DNS Resolver and upstream DNS servers.
- Test LAN, WAN, DNS, and Internet connectivity.
- Create and test a firewall rule to block Facebook related traffic.
- Setup of an Intrusion Detection System (IDS) using Snort within a pfSense firewall

System Requirements

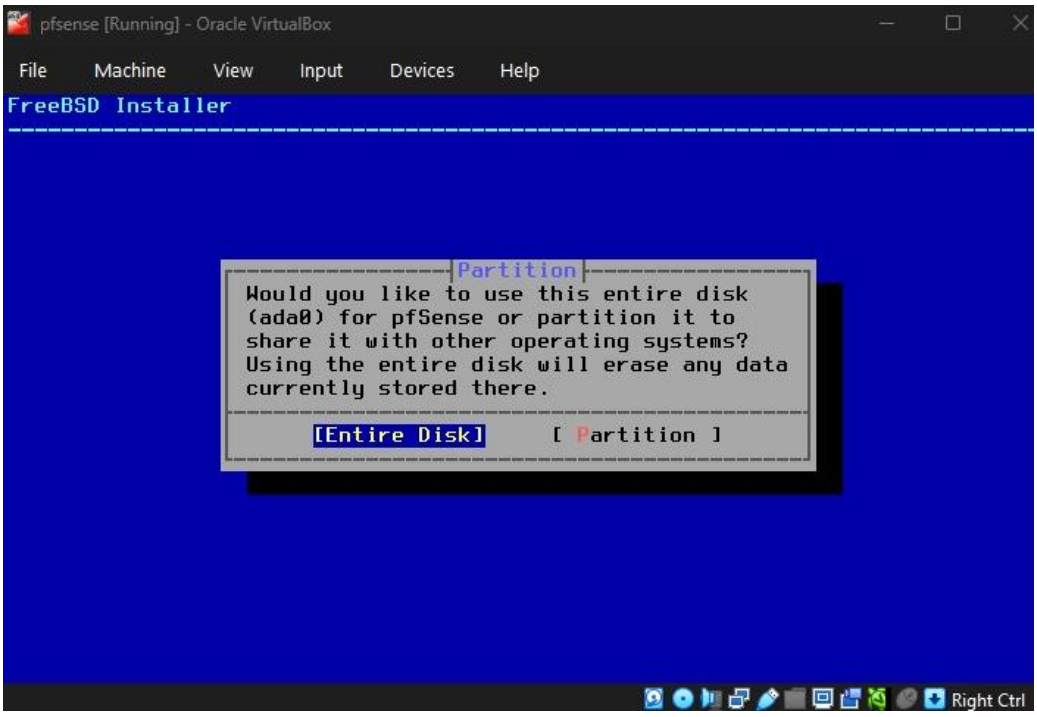
Item / Specification	Purpose
Host computer	Runs the hypervisor and both virtual machines.
pfSense VM - 2 vCPUs, 2 GB RAM, 10 GB disk, 2 NICs	Provides firewall, routing, DHCP, DNS, NAT, and security controls.
Windows client VM - 2 vCPUs, 2 GB RAM, 20 GB disk, 1 NIC Ubuntu client VM - 2 vCPUs, 2 GB RAM, 20 GB disk, 1 NIC	Acts as the internal LAN client for configuration and testing.
VirtualBox	Creates and connects the pfSense and Windows virtual machines.
pfSense ISO	Installs the pfSense firewall platform.
Windows 10 or Ubuntu	Installs the LAN client operating system.
Bridged WAN and Internal Network LAN	Separates WAN connectivity from the protected LAN.
Web browser, Command Prompt, pfSense Diagnostics	Used for configuration, verification, and troubleshooting.

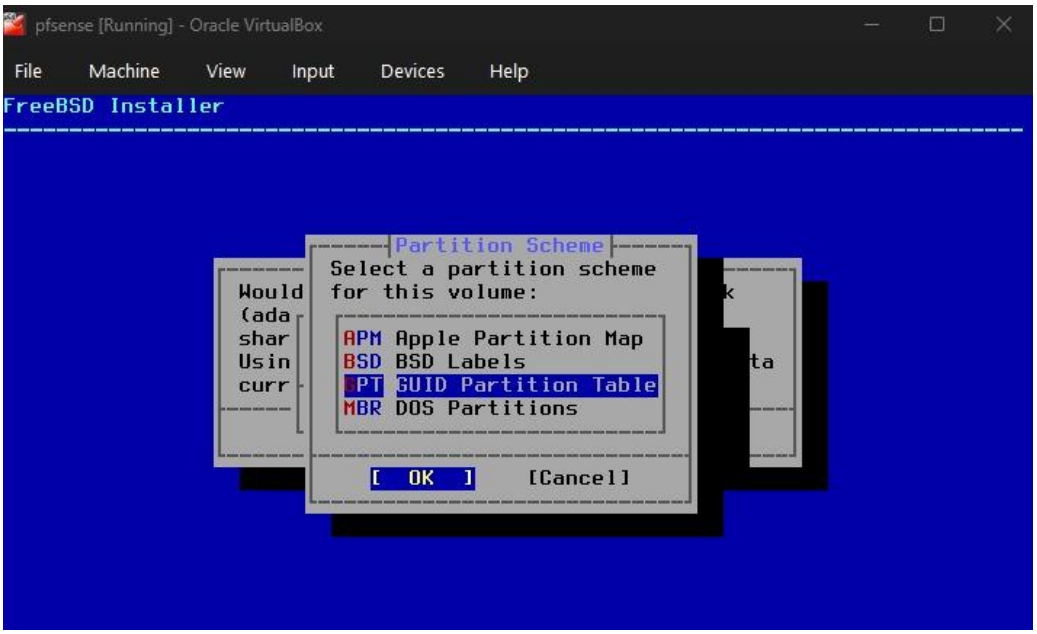
Network Topology

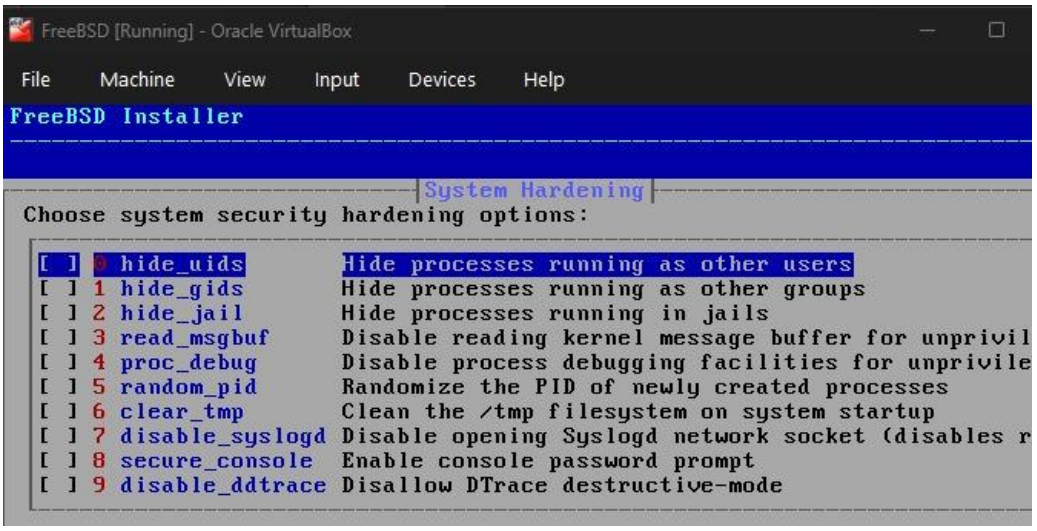
Network Element	Planned Configuration
pfSense WAN	10.3.8.166/24
pfSense LAN	10.1.10.1/24
Second Client Machine	10.1.10.10/24
Windows Default Gateway	10.1.10.1
DHCP Scope	10.1.10.100 - 10.1.10.200
Client DNS	10.1.10.1 through pfSense DNS Resolver
WAN Adapter Mode	Bridged
LAN Adapter Mode	Internal Network

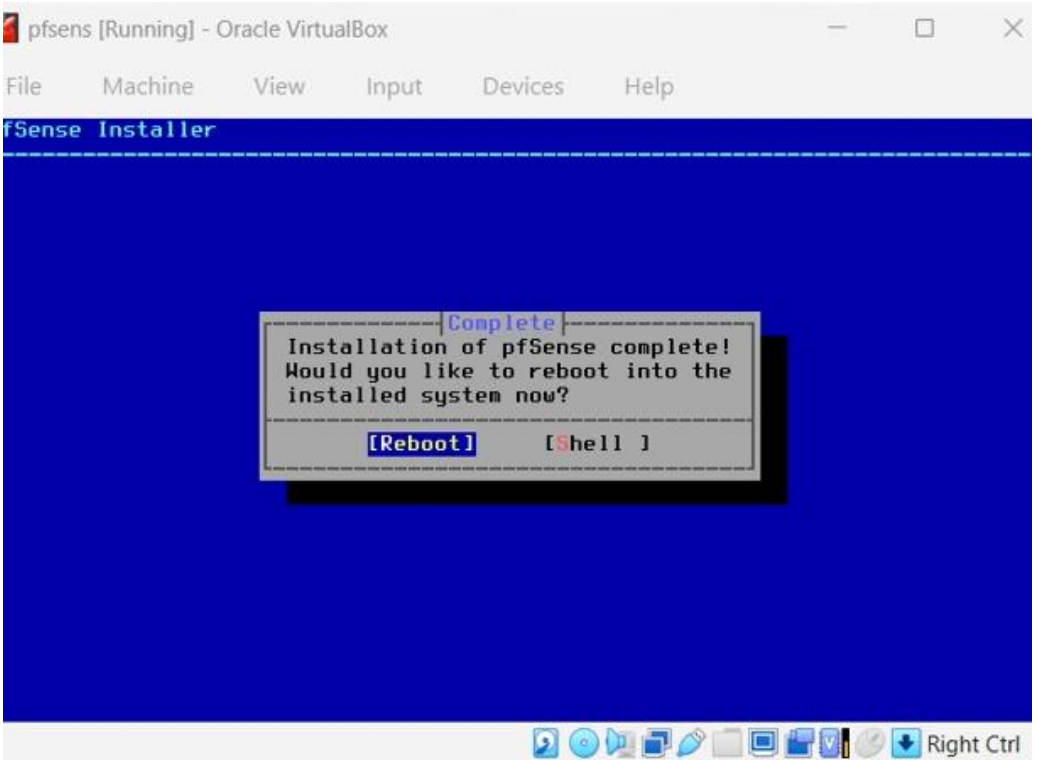
Installation procedure

1. Booted VM1 from the pfSense installation ISO attached as a virtual optical drive.
2. Selected "Install pfSense" from the boot menu to launch the installer.
3. Accepted the default keymap selection at the keymap configuration screen, as the lab used a standard US layout.
4. Selected "Auto (UFS) Guided Disk Setup" partitioning, allowing the installer to automatically partition and format the 10 GB virtual disk using the UFS filesystem.
5. Allowed the installer to copy the base system files and complete the installation.
6. At the post installation console menu, assigned em0 to the WAN interface and em1 to the LAN interface.




pfsense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
FreeBSD Installer
Partition Scheme
Select a partition scheme for this volume:
APM Apple Partition Map
BSD BSD Labels
GPT GUID Partition Table
MBR DOS Partitions
[OK] [Cancel]


FreeBSD [Running] - Oracle VirtualBox
File Machine View Input Devices Help
FreeBSD Installer
System Hardening
Choose system security hardening options:
[ ] 0 hide_uids Hide processes running as other users
[ ] 1 hide_gids Hide processes running as other groups
[ ] 2 hide_jail Hide processes running in jails
[ ] 3 read_msgbuf Disable reading kernel message buffer for unprivil
[ ] 4 proc_debug Disable process debugging facilities for unprivile
[ ] 5 random_pid Randomize the PID of newly created processes
[ ] 6 clear_tmp Clean the /tmp filesystem on system startup
[ ] 7 disable_syslogd Disable opening Syslogd network socket (disables r
[ ] 8 secure_console Enable console password prompt
[ ] 9 disable_ddtrace Disallow DTrace destructive-mode


pfsens [Running] - Oracle VirtualBox
File Machine View Input Devices Help
fSense Installer
Complete
Installation of pfSense complete!
Would you like to reboot into the installed system now?
[Reboot] [Shell]
Right Ctrl

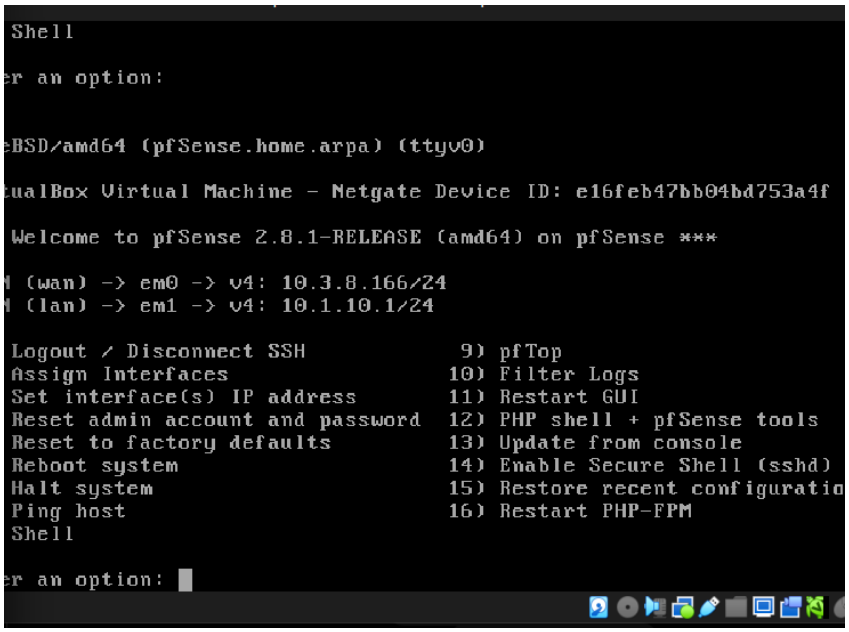
Configuration Procedure

WAN configuration

From the pfSense console menu Set interface IP address was selected to configure the WAN interface. A static IP address of 10.3.8.166 with a subnet mask of 255.255.255.0/24 was assigned to establish connectivity with the simulated external network.

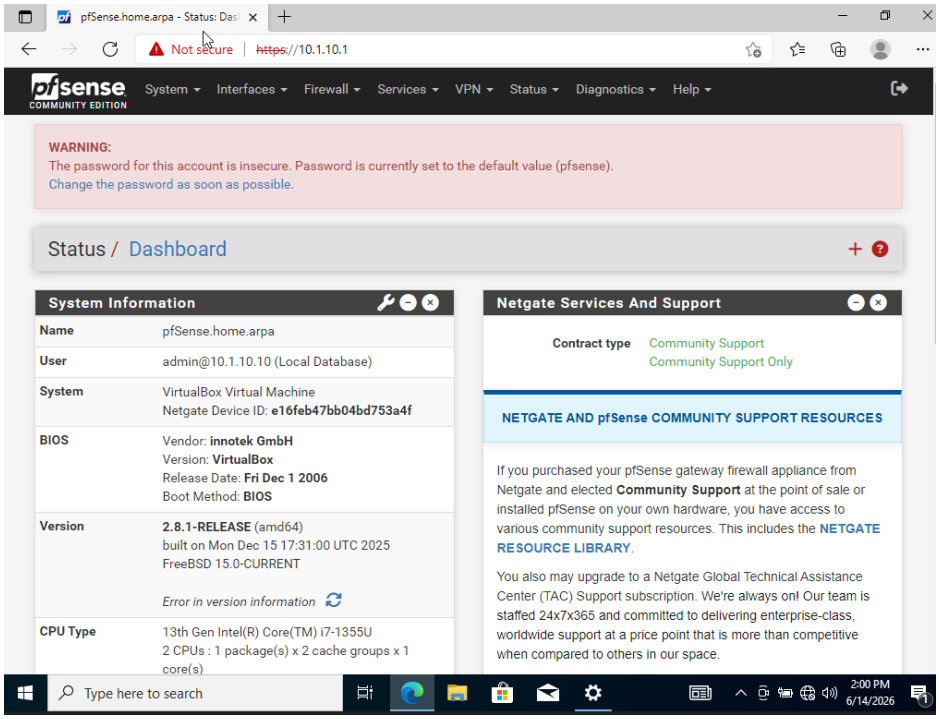
LAN Interface Configuration

The same console menu option was used to configure the LAN interface which was assigned the static address 10.1.10.1/24. This address serves as both the routed gateway for the LAN_Network segment and the Web GUI accessed.

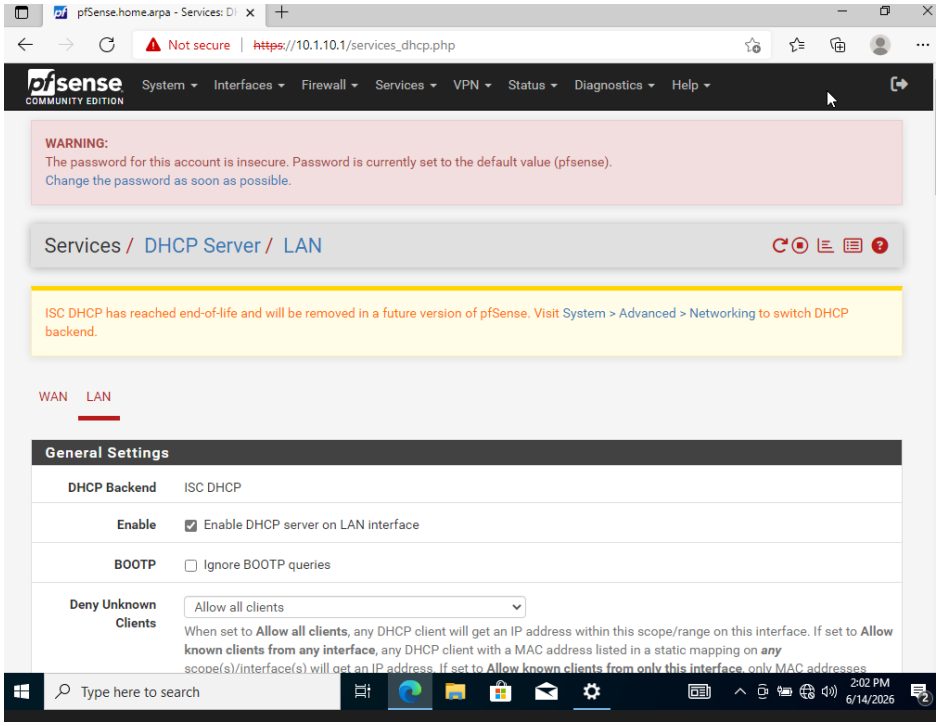


IP Addressing, DHCP

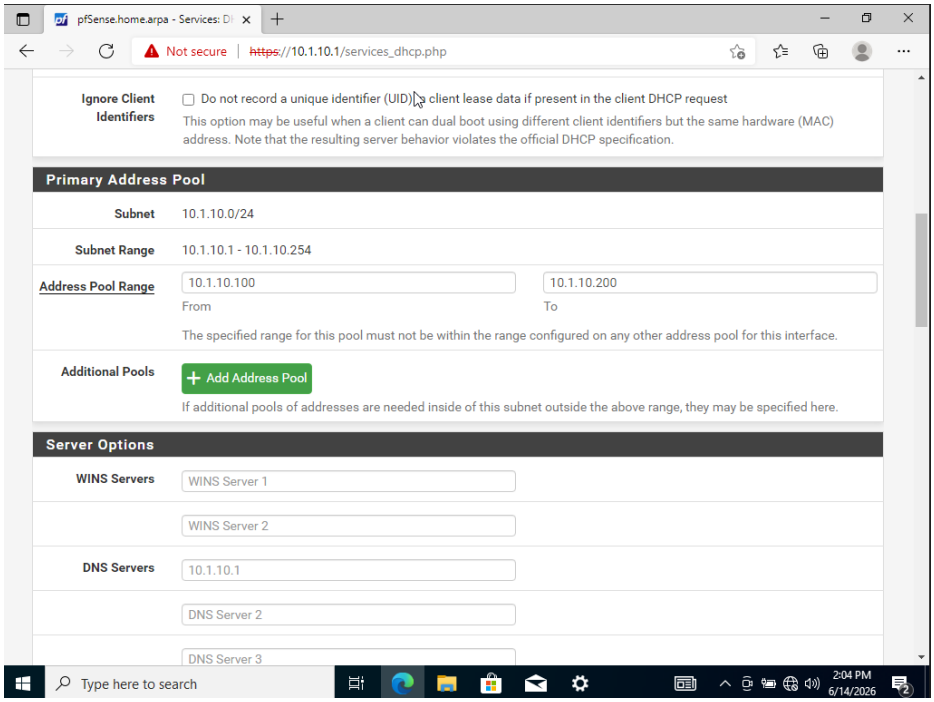
From VM2, a web browser was used to navigate to http://10.1.10.1 to log into the pfSense Dashboard



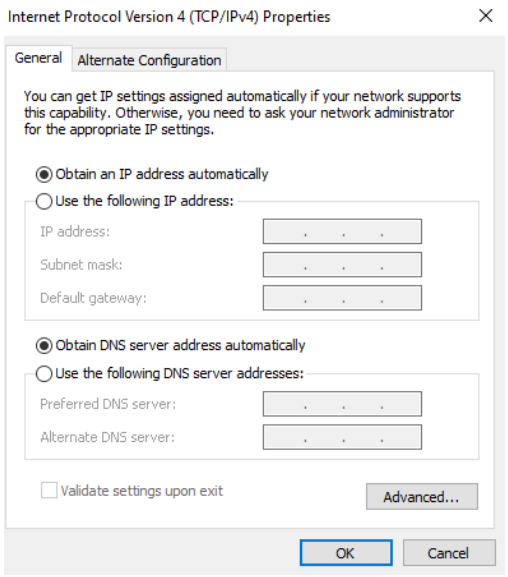
Navigated to Services > DHCP Server > LAN and enabled the DHCP server.



The Address Pool Range was configured from 10.1.10.100 to 10.1.10.200.

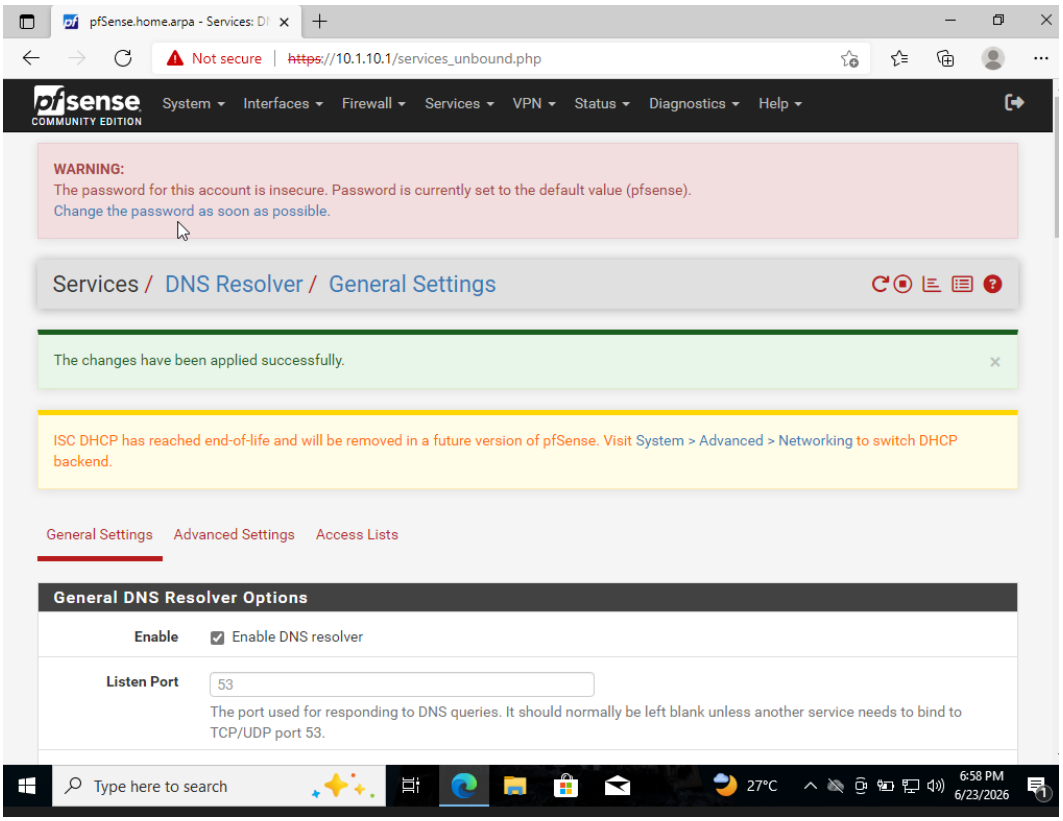


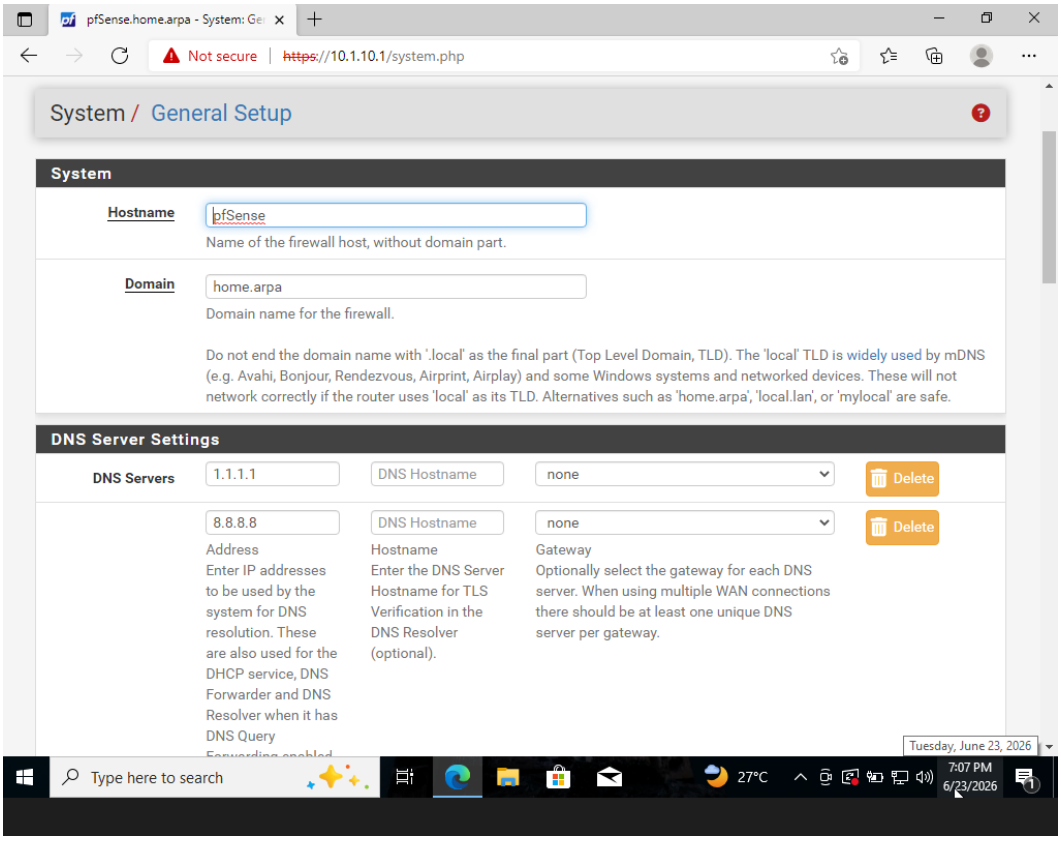
On VM2, the IPv4 properties were set to obtain an IP automatically.



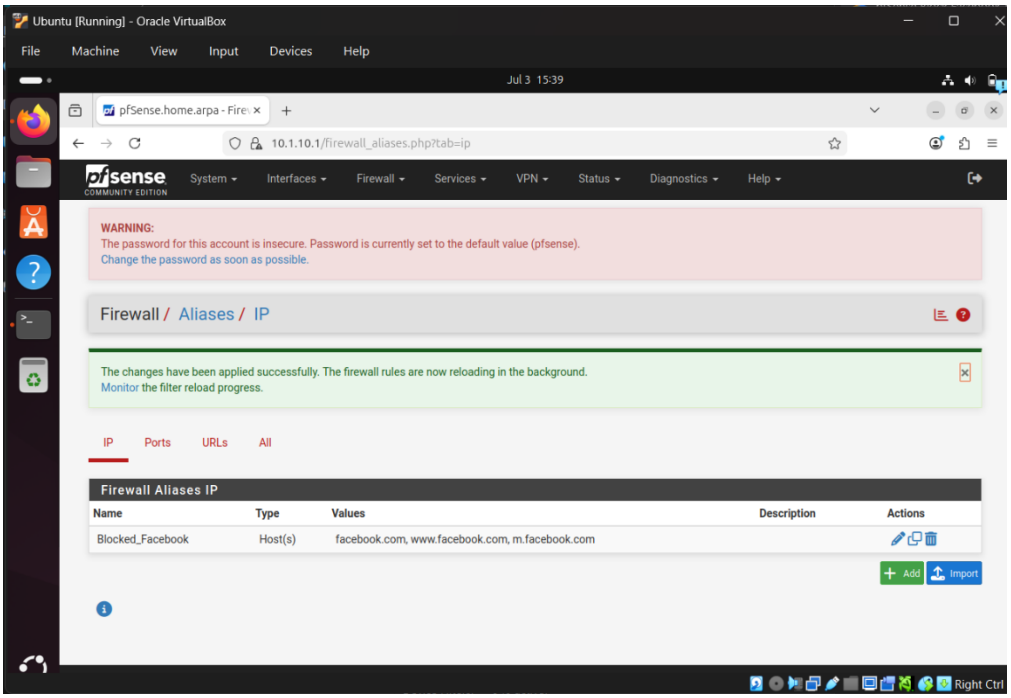
DNS Configuration

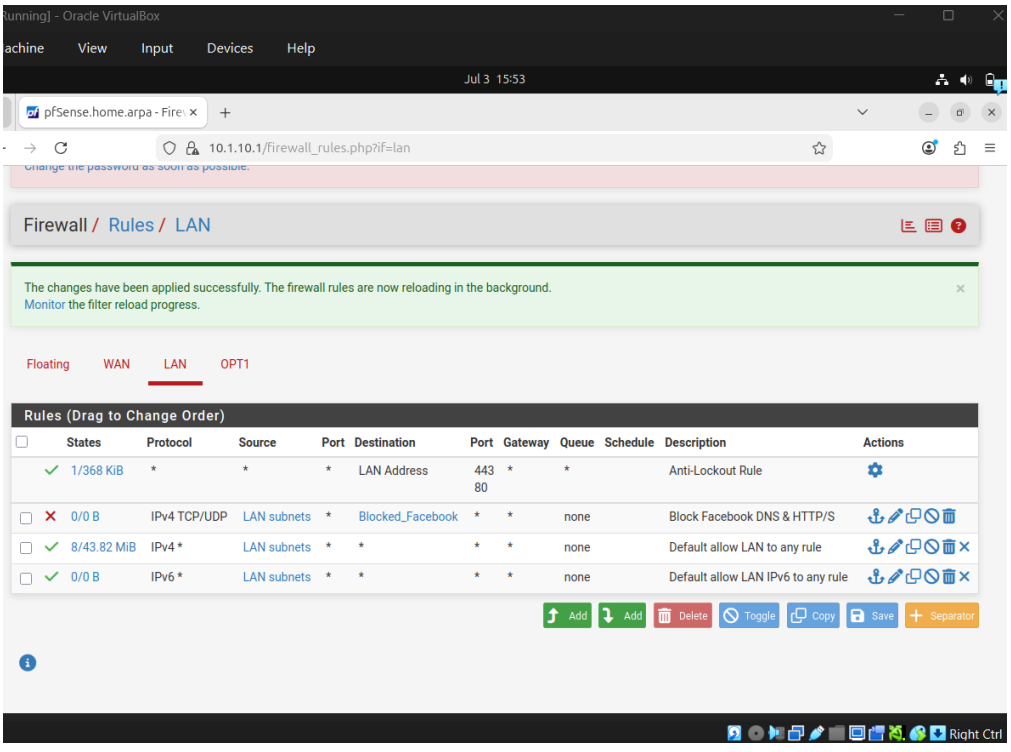
Accessed the pfSense Web GUI and navigated to Services → DNS Resolver.





1. Navigated to Firewall > Aliases and created a Host alias named Blocked_Facebook containing target domains (facebook.com, www.facebook.com, m.facebook.com, *.facebook.com).
2. Navigated to Firewall > Rules > LAN.
3. Created a new Block rule for TCP/UDP protocols originating from the LAN net, with the destination set to the Blocked_Facebook alias.
4. Moved the block rule above the default "Allow LAN to any" rule to ensure prioritization.

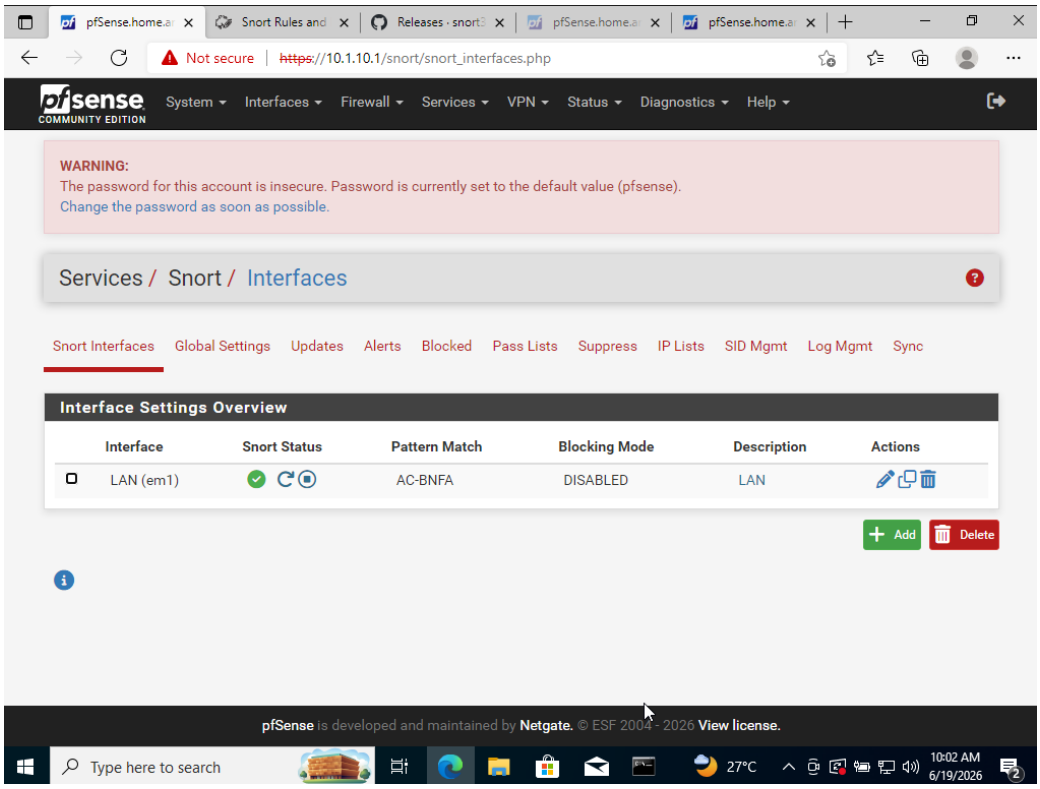


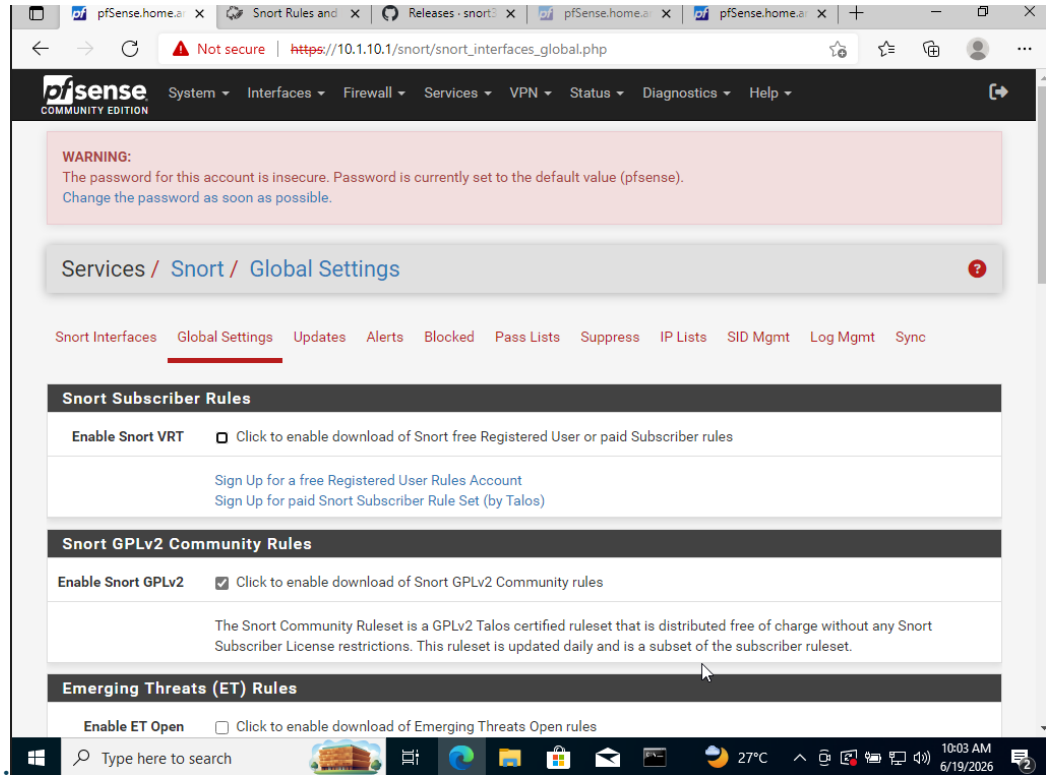


Network IDS Configuration using snort

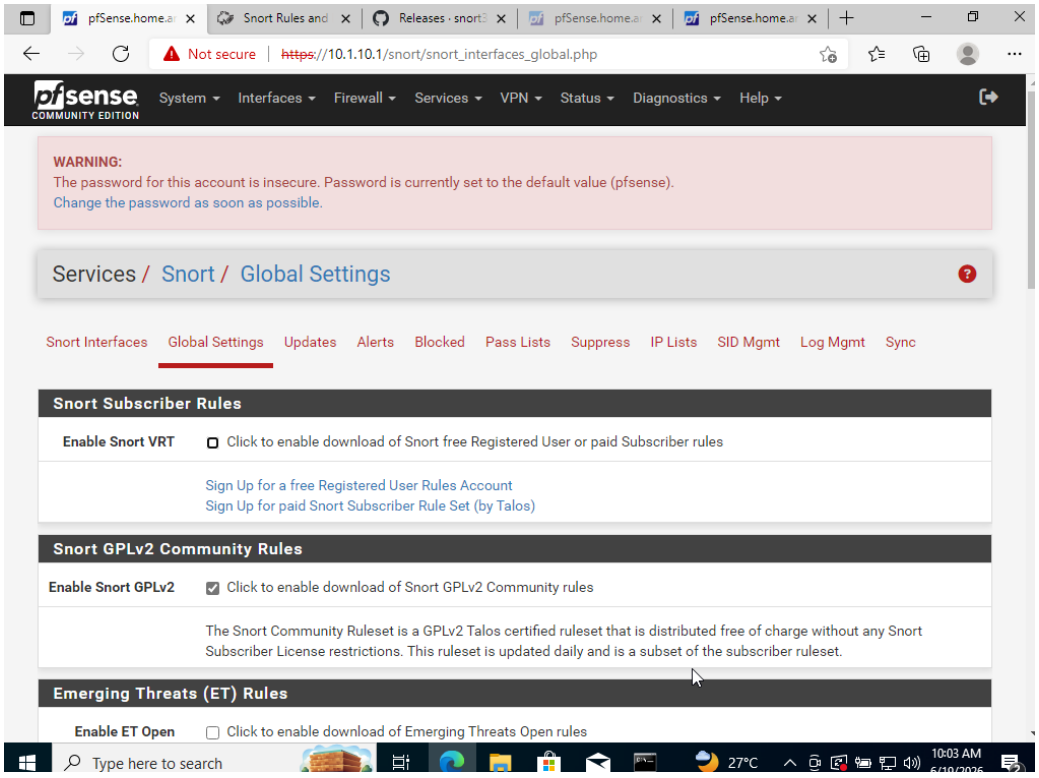
This demonstrates the practical setup and evaluation of a signature based Network Intrusion Detection System (IDS) using Snort within a pfSense firewall layout. Signature based intrusion detection identifies known threats by matching live infrastructure traffic sequences against pre defined rules.

Network Attribute	Assigned Parameter Value	Functional Role
pfSense Management IP	10.1.10.1	Default LAN Gateway & WebGUI Target
Host Testing Machine IP	10.1.10.100	Traffic Generator / Simulation Node
Snort Target Bound	LAN Interface (em1)	Monitored Broadcast Domain Segment





In the Snort engine settings dashboard open source community policies are turned on to download base signature components across the appliance.



Local laboratory signatures were crafted and written to the custom.rules definitions folder on the LAN Rules directory. Custom rules require specific formatting syntax fields to parse correctly a rule action, network protocol, network origin block, directional evaluation operators, destination macro block, and explicit rule option parameters enveloped in parentheses.

The following two specific rules were provisioned into the engine,

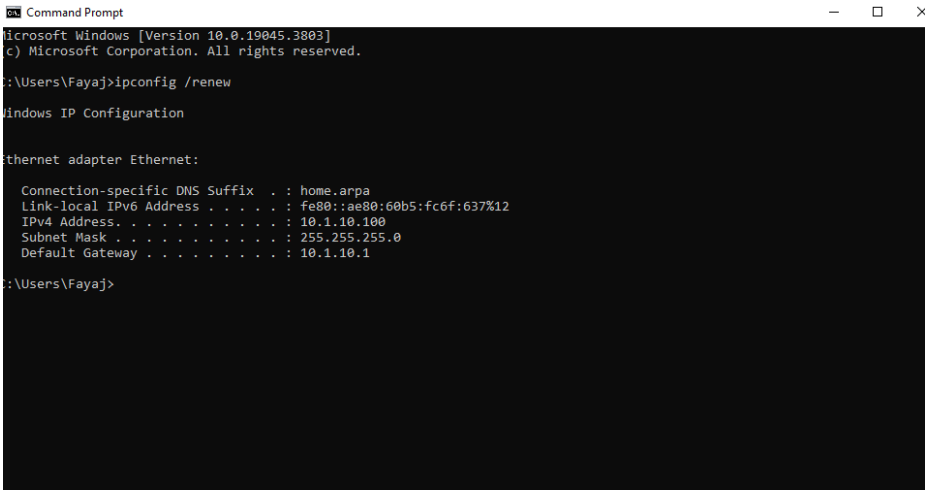
```
alert icmp any any -> $HOME_NET any (msg:"EDU TEST ICMP Ping detected"; sid:1000001; rev:1;)

alert tcp any any -> $HOME_NET 80 (msg:"EDU TEST HTTP Access detected"; sid:1000002; rev:1;)
```

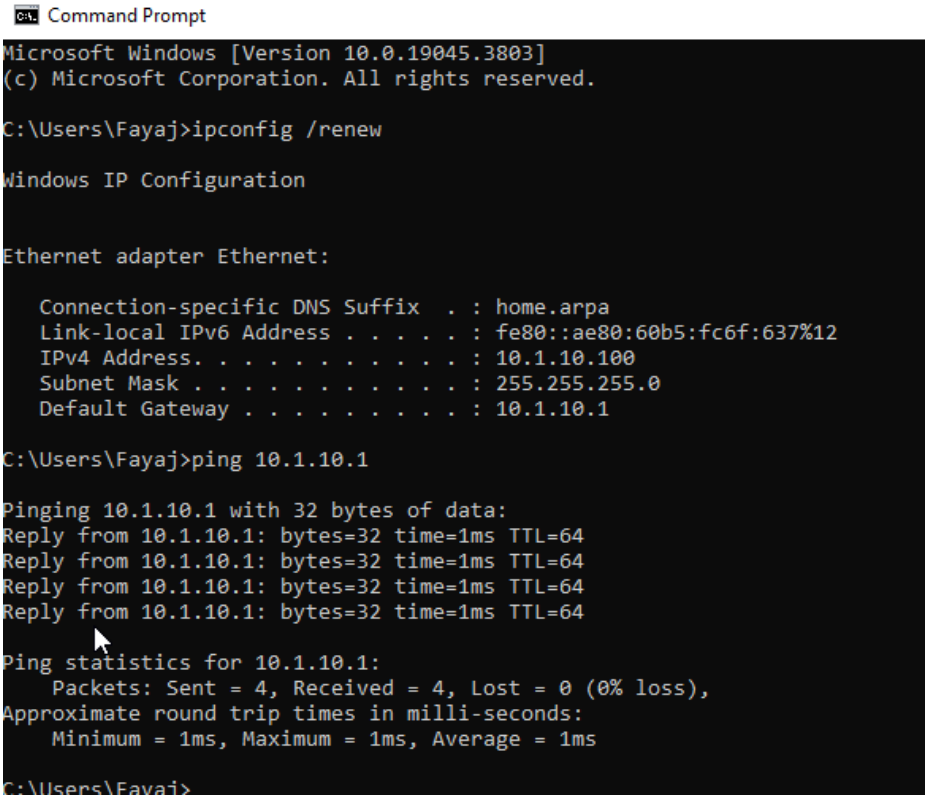
Testing and Results

Connectivity DHCP

Executed ipconfig /renew. The machine successfully received the IP 10.1.10.100.

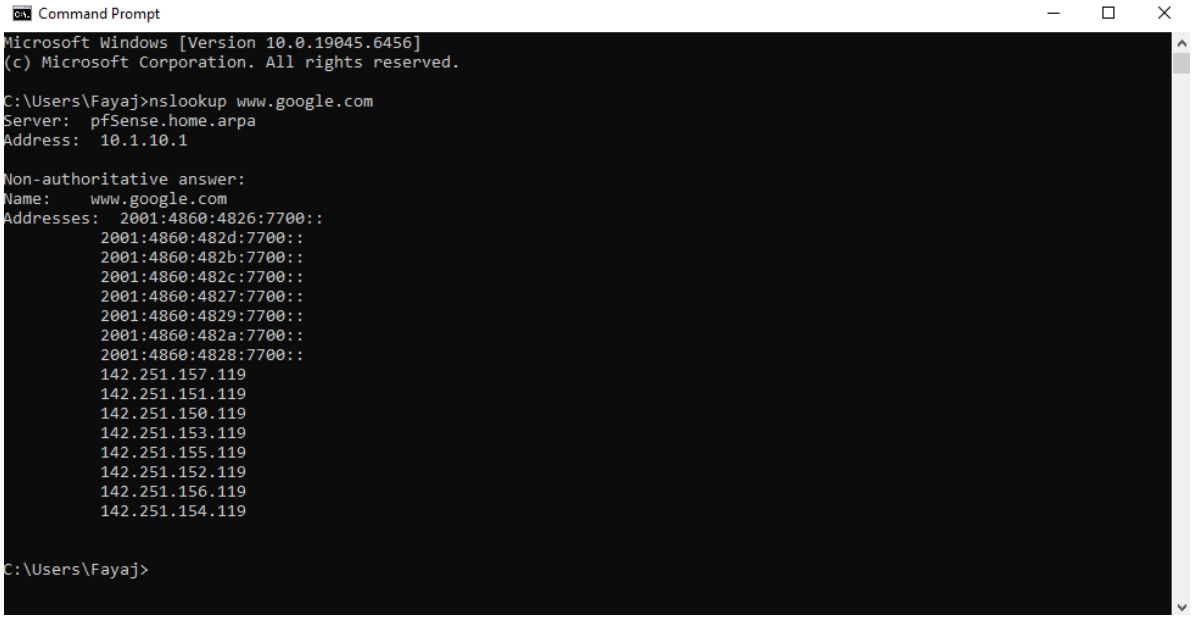


Executed ping 10.1.10.1 to confirm full two-way LAN communication.



DNS Configuration

Executed the command nslookup www.google.com within the Command Prompt



Internet Access Test

Results

```
PING 8.8.8.8 (8.8.8.8) from 192.168.100.15: 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=255 time=82.410 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=67.576 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=64.594 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 64.594/71.527/82.410/7.791 ms
```

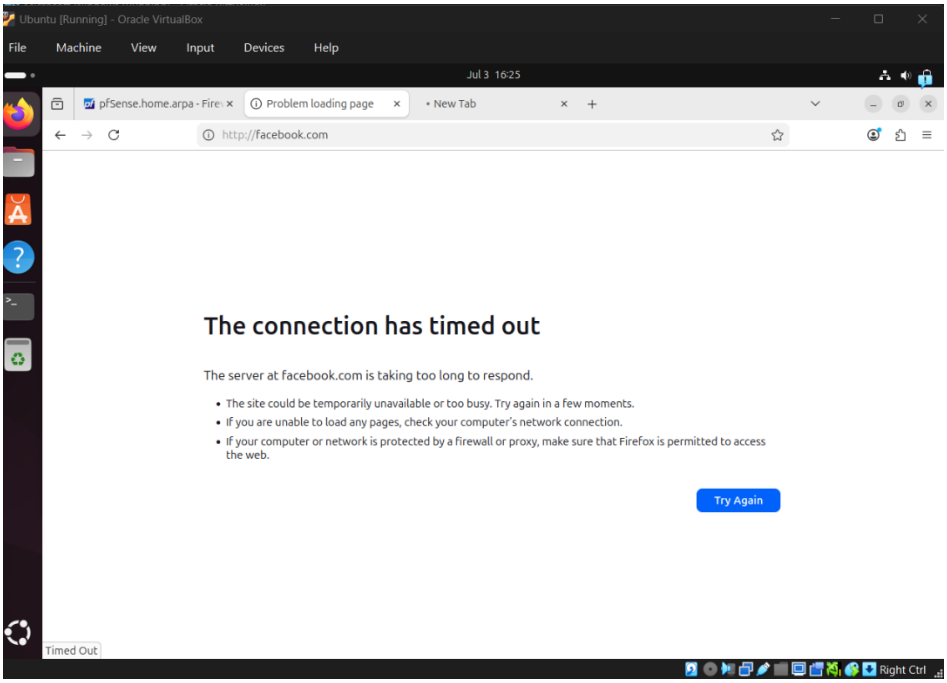
Results

```
PING 1.1.1.1 (1.1.1.1) from 10.1.10.1: 56 data bytes
64 bytes from 1.1.1.1: icmp_seq=0 ttl=255 time=462.615 ms
64 bytes from 1.1.1.1: icmp_seq=1 ttl=255 time=313.552 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=255 time=221.228 ms

--- 1.1.1.1 ping statistics ---
3 packets transmitted, 3 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 221.228/332.465/462.615/99.449 ms
```

Firewall Rule Enforcement Test

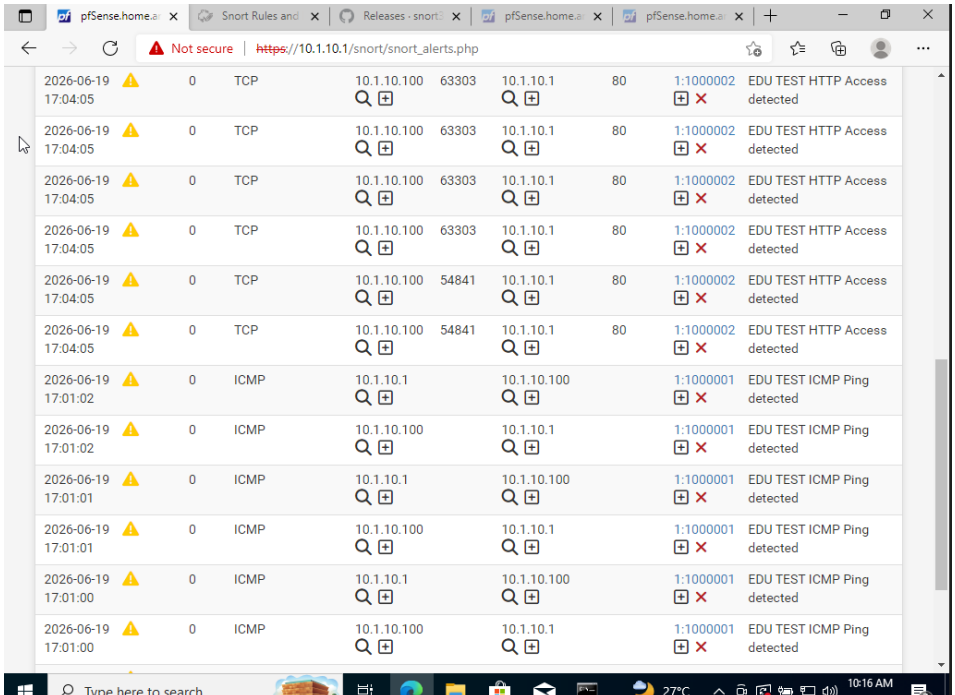
Facbook blocked using configurations



IDS Configuration using snort

The lab tests successfully verified that the Snort IDS engine can correctly inspect network layer data packets and trigger logging based on custom rule criteria.

This hands on work illustrates how custom local signature profiles can be targeted to flag anomalous or unsanctioned system traversal behaviors, providing a solid foundational model for complex corporate enterprise network security designs.



Issues and Solutions

Problem	Root Cause	Solution Implemented
nslookup facebook.com continued to resolve immediately after the block rule was applied.	The Windows client held a stale entry in its local DNS resolver cache from before the rule was active, so the query never reached the network layer where the block would apply.	Executed ipconfig /flushdns on VM2 to purge the local cache, forcing a fresh resolution/connection attempt that was then subject to the firewall rule.
Traffic to Blocked_Facebook was still permitted despite the block rule being present in the ruleset.	pfSense evaluates firewall rules top-down and applies the first match; the pre-existing default LAN allow rule was positioned above the new block rule, so it matched and permitted the traffic before the block rule was ever reached.	Manually re-ordered the block rule to the top of the LAN rule stack (Firewall > Rules > LAN), ensuring it is evaluated before the default allow rule.

Conclusion

Summary

This practical successfully demonstrated the end to end construction of a functional edge security gateway on virtualized infrastructure. Starting from a bare pfSense ISO, a dual interface firewall was installed, addressed, and configured to provide DHCP, DNS resolution, NAT, and policy based traffic filtering to an isolated internal network, with all core objectives verified through direct testing on a Windows client.

Knowledge and Skills Gained

- Virtual network segregation using VirtualBox Bridged and Internal Network adapter modes to model a realistic WAN/LAN boundary.
- Firewall rule evaluation logic, specifically the top down, first match behaviour that governs pfSense and the operational necessity of correct rule ordering.
- Interface provisioning and core service configuration, including static addressing, DHCP scope design, and DNS forwarder configuration.
- Practical experience with stateful packet inspection and log based verification as a method of confirming policy enforcement rather than relying on client-side behaviour alone.

References

Netgate pfSense Documentation

https://docs.netgate.com/pfsense/en/latest/

pfSense Community Edition

https://www.pfsense.org/